

Arun Harrish Andavar Padmasekar

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SUMMARY

Graduate student in Mechanical Engineering at Georgia Tech with expertise in FEA, CFD, Conjugate Heat Transfer (CHT), and surrogate modeling. Skilled in developing and validating computational and experimental models for complex systems, with proficiency in ANSYS, Fluent, GT-SUITE, MATLAB, and Python. Experienced in parametric design, DOE, optimization algorithms (NSGA-II, gradient-based), and machine learning (Gaussian Process Regression, regression models, meta-modeling and Neural Networks).

EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY

Master of Science in Mechanical Engineering

Cumulative GPA: 4.00/4.00

Atlanta, GA

Aug 2024 - Aug 2026 (Expected)

PSG COLLEGE OF TECHNOLOGY

Bachelor of Engineering in Mechanical Engineering

Cumulative GPA: 3.97/4.00

Coimbatore, India

Sept 2020 - April 2024

WORK EXPERIENCE

LIQUIDPISTON, INC.

Bloomfield, CT

Modeling and Simulation Intern

May - Aug 2025

- Conducted CHT on coolant jackets to identify hotspots and perform heat balance; validated pressure drop curves within 5% error
- Developed coolant jacket resistance models; proposed orifice restrictor plates, improving flow balance error from 37% to 17%
- Advanced pintle injector multiphase model from SAE 2023 work; optimized to <1% error in mass flow and <2% error in lift tracking
- Modeled 0D/1D rotor oil paths and coolant circuits; optimized friction multipliers to match pressure drop curves within 3% error
- Performed CHT analysis on cold start heater; proposed fin design changes that increased ΔT by 50% and boosted rating by 40%

GEORGIA INSTITUTE OF TECHNOLOGY – CRANE SAFETY RESEARCH CENTER

Atlanta, GA

Graduate Research Assistant

Jan 2025 - Present

- Developed parametric steady RANS CFD models to predict wind loads on tower cranes using atmospheric boundary layer inflow
- Automated large parametric CFD workflows via scripting and executed simulations on Georgia Tech PACE HPC clusters
- Designed a novel area-based decomposition method to quantify aerodynamic exposure levels within lattice structures
- Built and validated a reduced-order aerodynamic model correlating CFD loads with geometric parameters, achieving <3% error
- Authored an ASME FEDSM 2026 conference paper documenting CFD methodology and model reduction strategies

BULL MACHINES PVT. LTD.

Coimbatore, India

Intern – New Product Development

Jun 2023 - April 2024

- Led a 3-member team redesigning a hydraulic reservoir; achieved 50% volume reduction and 35% manufacturing cost savings
- Enhanced reservoir stability using CFD streamline visualization to minimize turbulence and ensure oil stability
- Validated new design through prototype testing; achieved 92% correlation with ANSYS Fluent simulations

PROJECTS

MULTI-OBJECTIVE OPTIMIZATION OF FORMULA SAE CHASSIS

Jan - April 2025

- Developed parametric CAD model with 21 design variables; applied FEA for torsional stiffness, stress, and deformation load cases
- Built Gaussian Process Regression surrogates, validated with cross-validation achieving $R^2 > 0.95$ and <3% relative error
- Applied NSGA-II and gradient-based optimization, generating Pareto fronts to quantify weight vs. stiffness trade-offs
- Achieved optimized chassis design 15–25% lighter while maintaining >7000 Nm/deg torsional stiffness

RELIABILITY ASSESSMENT OF HPC CHIP SOLDER JOINTS

March - April 2025

- Modeled 2D/2.5D semiconductor package in ANSYS with coupled thermal-structural analysis
- Performed cyclic thermal fatigue simulations; validated with mesh convergence and literature benchmarks
- Predicted solder joint life of ~10,000+ cycles under forced convection using Kanchanomai and Anderson models

RELIABILITY-BASED DESIGN OPTIMIZATION OF VEHICLE KNUCKLE

Aug - Dec 2024

- Optimized suspension knuckle weight by developing deterministic and stochastic optimization algorithms in MATLAB
- Generated surrogate models using ridge and lasso regression, reducing simulation time by 60 % and computational effort
- Validated surrogate models against ANSYS FEA with <4% error in stress/deformation predictions
- Achieved 38% weight reduction while ensuring structural integrity under fatigue, stress and deformation constraints

LEADERSHIP EXPERIENCE

TEAM BAJA BHAI

Coimbatore, India

Team Captain – College Baja SAE India Team

Aug 2021 - Jan 2024

- Led the design and development of a 4WD ATV, optimizing suspension and steering systems using advanced CAD/CAE tools
- Fostered collaboration among team members while leading the design optimization and vehicle integration process
- Guided a 25-member team, achieving innovative solutions and a top 10 finish at Baja SAE India among 80 teams

TECHNICAL SKILLS

- Tools:** SolidWorks, ANSYS, Fluent, COMSOL, OpenFOAM, ParaView, GT-SUITE, MATLAB, Python, Simulink, Creo Parametric, CATIA, Microsoft Office Suite
- Core Skills:** Design Optimization, Modeling and Simulation, CFD, CHT, FEA, Machine Learning, Numerical Heat Transfer